Vijay K

Bachelor of Engineering
Computer Science and Engineering
Chennai
+91 9361693537

Vijay

Objective

Computer Science undergraduate with experience building scalable backend systems, recommendation platforms, and distributed processing pipelines using Python, FastAPI, Redis, Celery, and Docker. Interested in AI-powered talent systems, large-scale data platforms, and software engineering.

Education

B.E. Computer Science and Engineering - Sri Sivasubramaniya Nadar College of Engineering - Chennai — 2023 – 27
CGPA: 8.35

Higher Secondary - Lakshmi Matric Hr. Sec. School - Trichy — 2022 – 23
Percentage: 97.8%

Experience

•**Labmentix** — *Remote*
Python Developer Trainee (Industry-Guided Internship Program) — June 2026 – Present

– Collaborated in a mentor-guided development environment to build secure document-signing workflows using FastAPI and Supabase.
– Participated in API design, code reviews, debugging, and iterative feature delivery following software engineering best practices.
– Contributed backend modules supporting authentication, PDF processing, and document lifecycle management.

Projects

SkillGraph: AI-Powered Workforce Intelligence Platform — *April 2026*
Python, FastAPI, PostgreSQL, Neo4j, Docker, Redis, Celery, FAISS, Next.js

– Designed and implemented a workforce intelligence platform using PostgreSQL and Neo4j to model 100+ entities and 8 relationship types, enabling multi-hop graph traversal and career pathway recommendations.
– Built asynchronous processing pipelines using Celery and Redis to offload embedding generation, resume parsing, and model updates from user-facing APIs, ensuring responsive request handling under concurrent workloads.
– Optimized backend performance by replacing sequential database operations with batched Cypher queries and caching precomputed embeddings, reducing worst-case API latency from 150s to under 2s (98.6% improvement) and lowering memory usage from >600MB to <120MB (80% reduction).
– Developed and deployed a containerized multi-service architecture using Docker, integrating FAISS-based semantic retrieval to support low-latency search and cloud-native deployment across Render and Hugging Face Spaces.

Document Signature Platform — *June 2026*
FastAPI, Next.js, PostgreSQL/Supabase, JWT

– Designed and implemented a secure document-signing platform enabling authenticated users to upload, share, and digitally sign PDF documents through tokenized access workflows.
– Developed 20 REST API endpoints supporting authentication, document lifecycle management, public signing, document retrieval, and role-based access control using FastAPI and JWT.
– Built PDF processing pipelines to automate signature placement and signed-document generation for multi-page documents, supporting two distinct workflows: authenticated self-signing and tokenized guest signing with dynamic viewport-to-canvas coordinate mapping.
– Implemented structured validation, exception handling, and production-oriented backend practices to improve reliability and maintainability; deployed the application to cloud environments with environment-based configuration and secure service integration.

Open Source Experience

•**GirlScript Summer of Code (GSSoC) 2026** — *Remote*
Open Source Contributor — May 2026 – Present

– Contributed feature enhancements and bug fixes through GitHub-based collaborative development workflows involving issue discussions, code reviews, and pull request iterations.
– Successfully merged 2 pull requests and actively contributing to additional open-source issues.

Technical Skills

- **Languages**: Python, Java, JavaScript, SQL
- **Frontend Development**: React.js, Next.js
- **Backend & APIs**: FastAPI, Node.js, REST APIs, JWT Authentication
- **Databases & Caching**: PostgreSQL, Supabase, Neo4j, MongoDB, Redis
- **DevOps & Tools**: Docker, Git, GitHub Actions, Celery, CI/CD Pipelines
- **Machine Learning**: Scikit-learn, XGBoost, Random Forest, FAISS
- **Software Engineering**: Debugging, Unit Testing, API Design, OOP, Software Design
- **Algorithms**: Graph Algorithms, Dynamic Programming, Data Structures
- **Deployment**: Vercel, Render, Hugging Face Spaces

Hackathons and Achievements

•**Leetcode** — *Ongoing*
– Solved 200+ problems covering arrays, graphs, trees, dynamic programming, greedy algorithms, and binary search.

•**Construction Cost Prediction Challenge – Top 16% Finish** — *March, 2026*
– Ranked 80th out of 500+ participants by developing and evaluating XGBoost and Random Forest models for construction cost estimation and predictive analysis.

•**SAP HackFest 2025 – National Pre-Finalist** — *April, 2025*
– Advanced through 3 competitive evaluation rounds among national-level participating teams.

Positions of Responsibility

•**Head Volunteer** - Invente Paper Presentation — *2025*